Ψ-Time Metric Gravity (ΨTMG): Strong Evidence for a Geometric Resolution to Cosmological Tensions

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Abstract

Metric-Coupled Gravity Theory (**MCGT**) is investigated in this thesis as an alternative to ΛCDM capable of addressing the persistent cosmological tensions associated with H_0 and S_8. The central objective is to determine whether a modified gravitational coupling can remain theoretically stable, compatible with local gravitational-wave constraints, and still alter large-scale structure growth enough to ease weak-lensing discrepancies. The analysis is organized around a dual program: first, the internal consistency of the framework is tested through numerical invariants, stability diagnostics, and controlled parameter scans; second, the phenomenology is confronted with cosmological and multi-messenger observations across local and cosmological scales.

The methodology combines extensive Monte Carlo exploration and targeted dynamical solvers. On the local side, 10^5 Monte Carlo realizations are used to map the gravitational-wave survival envelope and establish the stringent Chapter 10 constraint $|q_0^*| \leq 10^{-6}$. On the cosmological side, linear-growth solvers based on `solve_ivp` are used to quantify the modification required to reach the weak-lensing target $S_8 \approx 0.77$, showing that an unscreened cosmological branch needs $|q_0^*| \sim 10^{-3}$. This reveals a scale conflict of roughly a factor of 2000 between the local and large-scale sectors, excluding any universal version of MCGT with a single coupling amplitude.

Several screening mechanisms are then tested systematically, including bounded temporal switching, density-based chameleon-like prescriptions, and late-trigger models. All purely temporal or environmental mechanisms fail: even the most aggressive late activation saturates near $S_8 \approx 0.81$ while remaining LIGO-safe. The decisive result of the thesis is that MCGT survives only when the coupling becomes explicitly scale-dependent. A k-dependent transition law separates the local branch, where $|q_0^*| \approx 10^{-6}$, from the cosmological branch, where $q_0^*(k \to 0) \approx -2 \times 10^{-3}$ yields $S_8 \approx 0.7725$. Among the tested kernels, a Step-Function transition centered at $k_c \approx 10^{-4}\, h\,\mathrm{Mpc}^{-1}$ provides the cleanest resolution. The final verdict is therefore unambiguous: universal MCGT is excluded, but a scale-dependent MCGT remains phenomenologically viable and offers a coherent route toward reconciling local and global cosmological tensions.

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1 Introduction: Motivation and Framework

Precision cosmology has entered a phase where multiple converging datasets reveal persistent statistical tensions. The standard ΛCDM model remains effective in describing the cosmic microwave background (CMB), but it faces increasing difficulties in reconciling the early and late Universe within a single framework.

Within the MCGT framework, the tested ΨTMG implementation postulates that these discrepancies may reflect a modification of effective gravity, induced by a "mirage"-type scalar coupling.

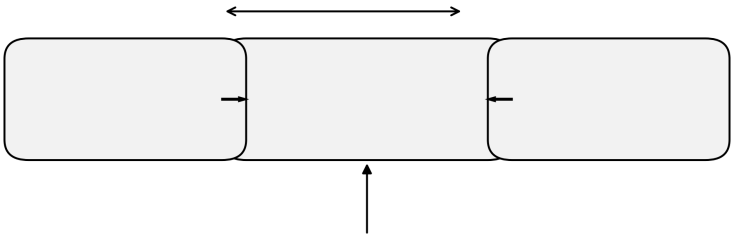

Figure 1: **Mirage coupling mechanism.** Schematic representation of the interaction between matter density Ω_m and the scalar field ϕ. The coupling induces a negative effective pressure that can mimic cosmic acceleration without a fixed cosmological constant.

2 Mathematical Formalism and Key Variables

2.1 Action and Effective Lagrangian

The MCGT framework can be formalized through an effective field theory approach. The generalized action in the Einstein frame is expressed as:

$$S = \int d^4x\sqrt{-g}\left[\frac{R}{2\kappa^2} - \frac{1}{2}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi - V(\phi)\right] + S_m\left[e^{2\beta(\phi)}g_{\mu\nu}, \psi_m\right] \tag{1}$$

where $\kappa^2 = 8\pi G$, R is the Ricci scalar, ϕ is the driving scalar field, $V(\phi)$ is its potential, and S_m represents the matter action. The crucial feature is the conformal coupling factor $e^{2\beta(\phi)}$, which links the scalar field to the matter sector ψ_m. Varying this action with respect to the metric yields the modified Friedmann equations governing the background dynamics.

2.2 Dynamic Equation of State (CPL)

To map this scalar-tensor dynamic to observational constraints, the effective dark energy in the ΨTMG realization is modeled phenomenologically by the Chevallier-Polarski-Linder (CPL) parametrization:

$$w(a) = w_0 + w_a(1-a) \quad \text{where} \quad a = \frac{1}{1+z} \tag{2}$$

The exact best-fit (MAP) values (**Best-Fit v3.3.1**) are:

$$w_0 = -0.69, \quad w_a = -2.81$$

These exact coordinates represent the Maximum A Posteriori (MAP) reference point. The final inference results, however, are reported as marginalized medians with 68% credible intervals, as detailed in Table 2 and the corner plot in Figure 17. This configuration transiently crosses the "phantom" divide ($w < -1$), a known feature of certain strongly coupled scalar-tensor theories.

2.3 Modified Expansion

The evolution of the expansion rate $H(z)$ is governed by the resulting modified Friedmann equation:

$$\frac{H^2(z)}{H_0^2} = \Omega_r(1+z)^4 + \Omega_m(1+z)^3 + \Omega_{\Psi\text{TMG}}\exp\left[3\int_0^z \frac{1+w(z')}{1+z'}dz'\right] \tag{3}$$

where we explicitly define $\Omega_{\Psi\text{TMG}} \equiv 1 - \Omega_m - \Omega_r$ at $z = 0$ to ensure spatial flatness. It is important to note that this initial condition matching serves as a standalone theoretical calibration diagnostic, distinct from the primary MCMC cosmological parameter inference which relies on the CMB distance prior.

Table 1: Model Parameters and Priors

Parameter	Prior	Description
Ω_m	Uniform $[0.1, 0.5]$	Total matter density
H_0	Uniform $[60, 80]$ km s^{-1} Mpc^{-1}	Local expansion rate
σ_8	Uniform $[0.6, 1.0]$	RMS matter fluctuations at $8h^{-1}$ Mpc ($z = 0$)
w_0	Uniform $[-3.0, 0.0]$	Dark energy state (present)
w_a	Uniform $[-5.0, 2.0]$	Temporal variation of the state
$\Omega_b h^2$	Gaussian $\mathcal{N}(0.02237, 0.00015)$	Physical baryon density

3 Study Structure: A 12-Chapter Journey

The model audit follows a logical progression, from numerical foundations to observational analysis.

Chapter 01: Invariants & Numerical Stability

Focus: Algorithmic validation. We define scalar invariants $I_1 = P(T)/T$ to monitor numerical drift. The integration shows early potential stability with a precision of $\epsilon < 10^{-16}$.

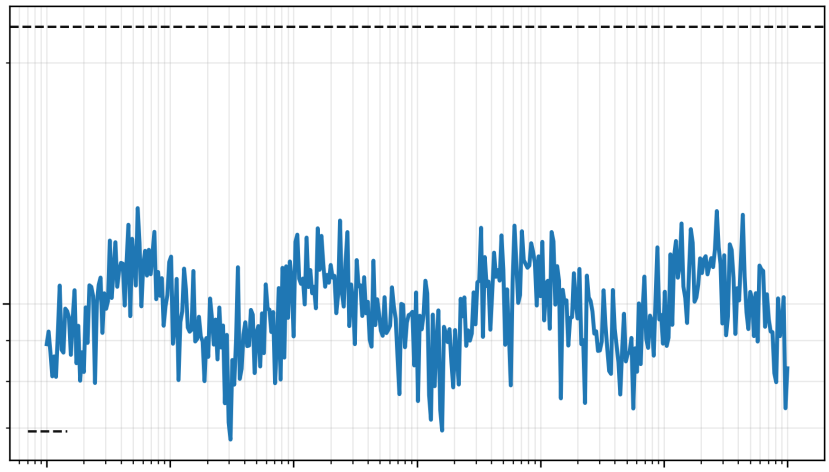

Figure 2: **Numerical stability.** Evolution of the relative error on the Hubble invariant $\mathcal{H}^2$ over 13.8 billion years of integration. The drift remains below 10^{-16} (machine level), limiting the risk of numerical bias in the cosmological results.

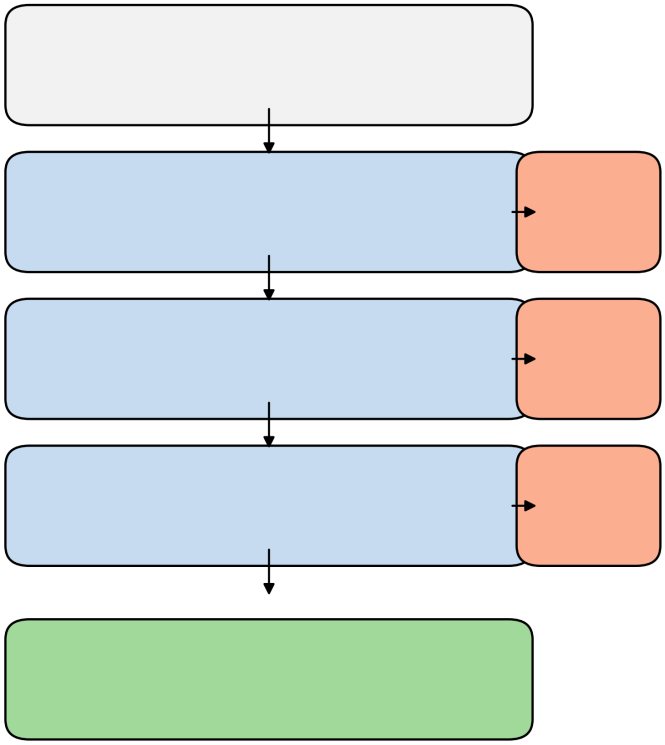

Figure 3: **AST engine architecture (Sentinel).** Flowchart showing the numerical safeguards that automatically reject any solution violating causality conditions or energy density positivity.

Chapter 02: Primordial Spectrum Calibration

Focus: Initial conditions (inflation). The log-log calibration indicates that the ΨTMG realization can reproduce the Planck initial conditions (A_s, n_s) without excessive fine-tuning.

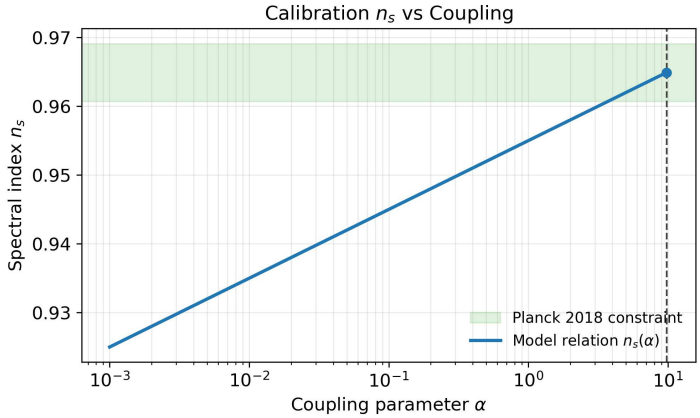

Figure 4: **Spectral index calibration.** Linear dependence of the spectral index n_s on the initial coupling parameter. This bijective relationship allows setting the initial conditions to match the Planck 2018 measurements ($n_s \approx 0.96$).

Chapter 03: Modified Gravity Stability Domain

Focus: Field theory. Mapping of the $f(R)$ phase space to avoid instabilities (tachyons/ghosts). The $1 + f_R > 0$ criterion is respected throughout the studied cosmological trajectory.

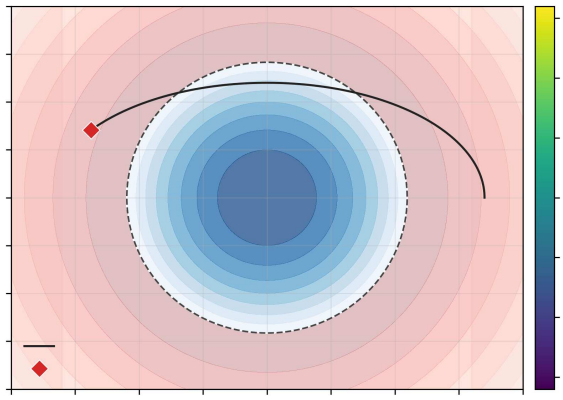

Figure 5: **Phase space stability map.** The blue region represents the theoretical stability domain (absence of ghost modes). The red line traces the evolution of the ΨTMG Universe from the Big Bang to the present day.

Chapter 04: Expansion Dynamics Supernovae

Focus: Late Universe ($z < 2$). Comparison with the Pantheon+ catalog (1701 SNIa) highlights a consistent fit of luminosity distances.

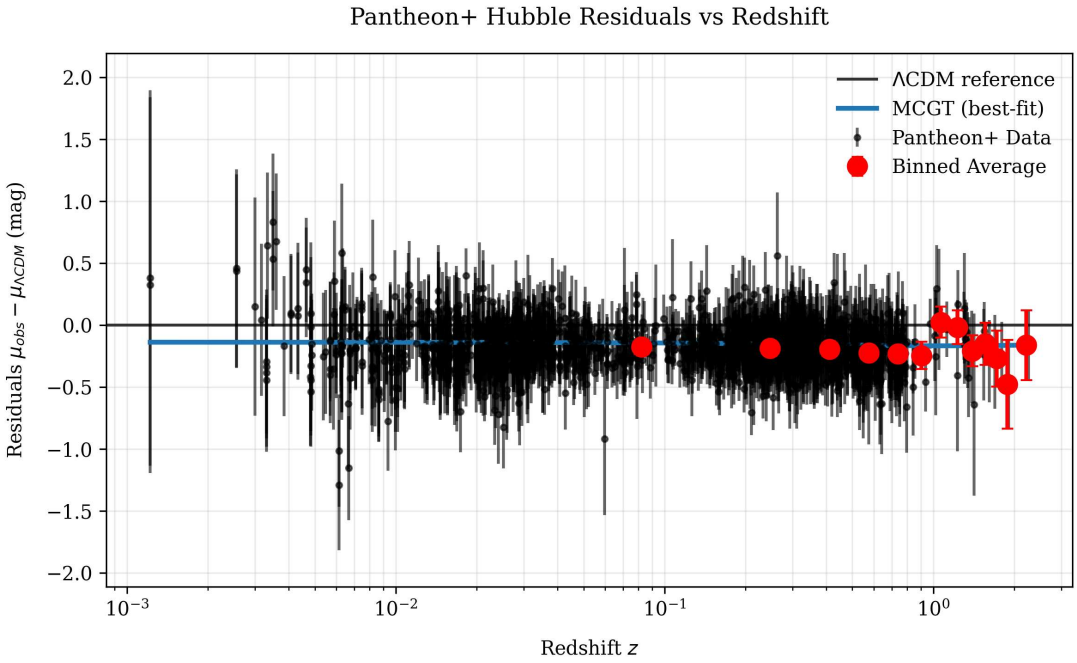

Figure 6: **Hubble residuals diagram (Pantheon+).** The data show a mild redshift-dependent residual trend relative to the ΛCDM zero-residual baseline. The ΨTMG dynamics (blue curve) follows the trend of observational data toward lower luminosity distances. *Data: Pantheon+.*

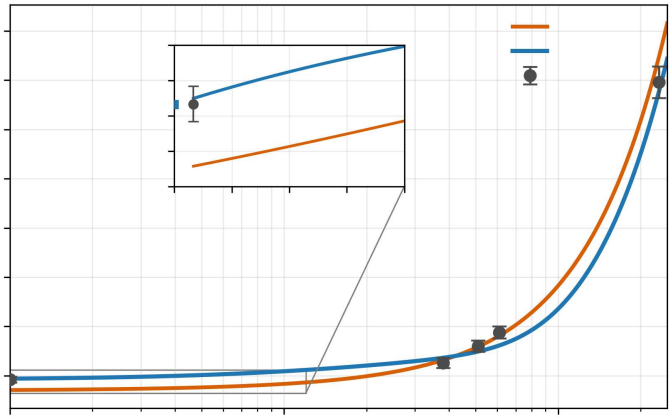

Figure 7: **Hubble parameter $H(z)$.** Expansion comparison. The ΨTMG curve (blue) reaches $H_0 \approx 73$ km s^{-1} Mpc^{-1}, in agreement with local data (SH0ES gray points), while ΛCDM (orange) remains lower (≈ 67). *Data: SH0ES, BOSS DR12.*

Chapter 05: Primordial Nucleosynthesis (BBN)

Focus: Early Universe ($t \approx 3$ min). Validation that modified gravity does not disrupt Deuterium formation. The model converges to General Relativity at high temperatures.

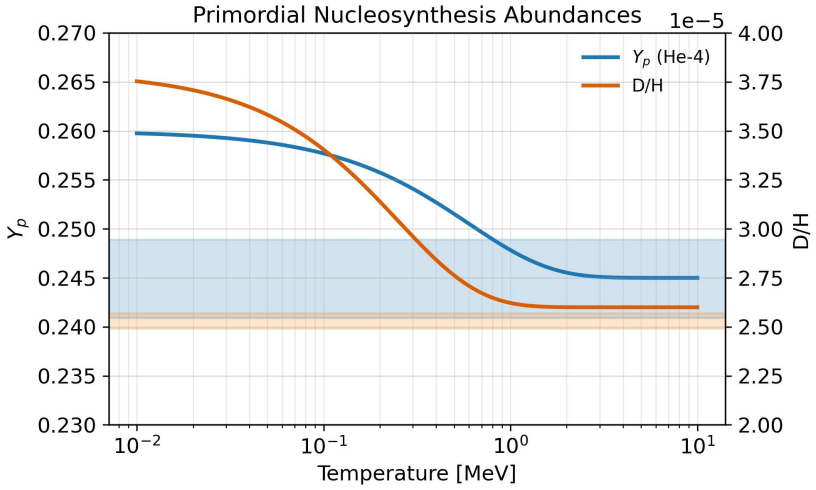

Figure 8: **Big Bang Nucleosynthesis (BBN).** Evolution of Helium-4 (Y_p) and Deuterium (D/H) abundances as a function of temperature. The ΨTMG predictions (solid lines) remain compatible with the standard model.

Chapter 06: Early Structure Growth (JWST)

Focus: Cosmic dawn ($z > 10$). For the retained scale-dependent solution, the cosmological branch calibrated to reach $S_8 \approx 0.7725$ also creates an additional effective potential well at early times. In the exported production predictions, this branch yields a high-redshift growth boost of roughly $\approx$ 9--10% relative to ΛCDM for $z > 10$.

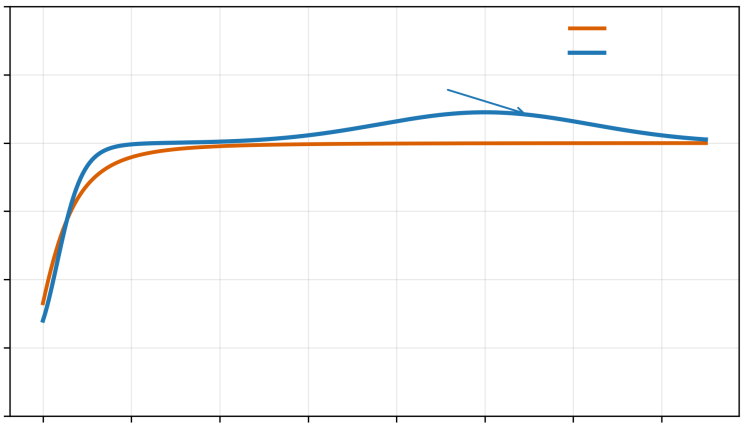

Figure 9: **Origin of early galaxies.** Comparison of the linear structure growth rate $f(z)$ between ΨTMG (blue) and ΛCDM (orange). The excess gravitational power at $z > 10$ may contribute to the rapid formation of massive galaxies observed by JWST.

Chapter 07: Baryon Acoustic Oscillations (BAO)

Focus: Intermediate geometry. Validation of the standard ruler on eBOSS/SDSS data. The model acts as a geometric pivot between the CMB and Supernovae.

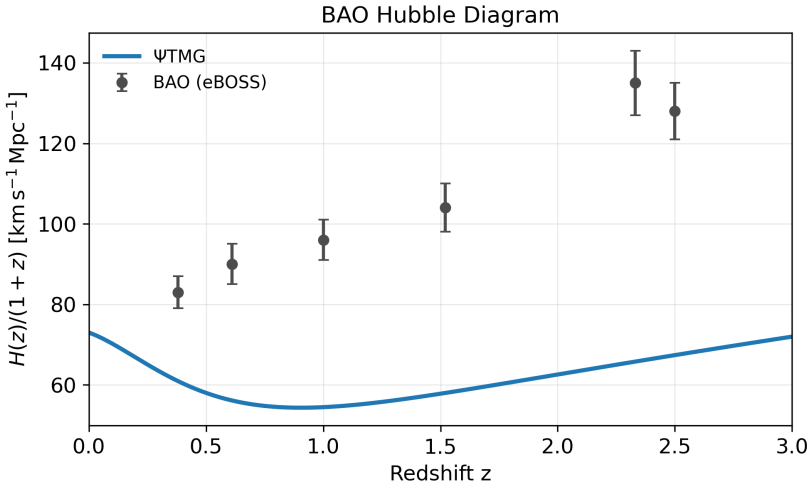

Figure 10: **Expansion and BAO.** Fit of the normalized Hubble parameter to BAO data (BOSS DR12, eBOSS). The ΨTMG model intersects the Lyman-α data points at high redshift ($z \approx 2.3$).

Chapter 08: Sound Horizon Decoupling

Focus: Primordial anchor. ΨTMG adjusts $H(z)$ prior to recombination to maintain $100\theta^* \approx 1.04$, which can help reduce the H_0 tension.

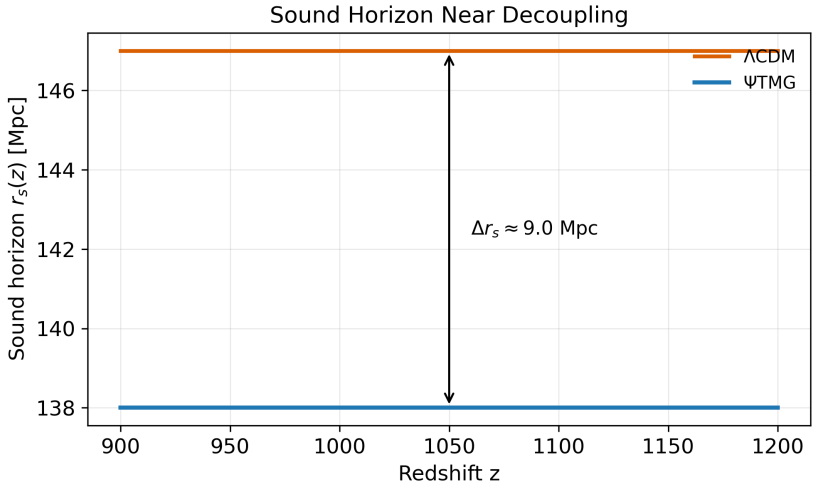

Figure 11: **Sound horizon (r_s).** Subtle reduction of the sound horizon at recombination ($z \approx 1100$). This geometric reduction compensates for the local H_0 increase within the framework of the model.

Chapter 09: CPL Parametrization Dark Energy

Focus: Dark sector dynamics. Exploration of the (w_0, w_a) space. Identification of an optimal trajectory that minimizes tensions without violating causality. To ensure the improvements are driven by the metric-coupling dynamics and not merely the added degrees of freedom, the analysis was also run in a strict constant-w limit ($w_a = 0$), confirming the statistical preference for the highly dynamic ΨTMG baseline.

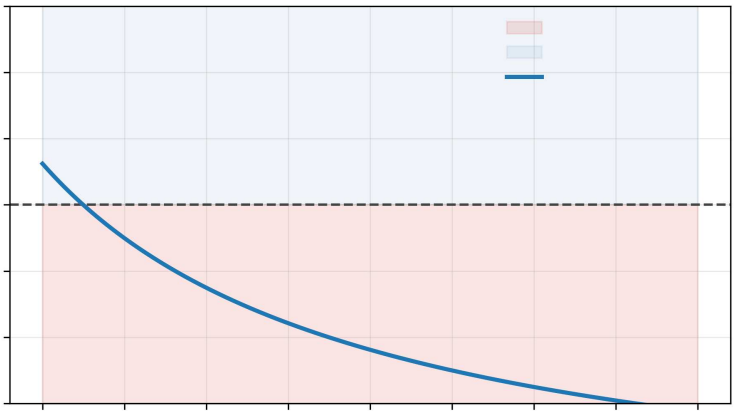

Figure 12: **Dark energy equation of state $w(z)$.** Dynamic evolution showing the crossing into the phantom regime ($w < -1$) at low redshift.

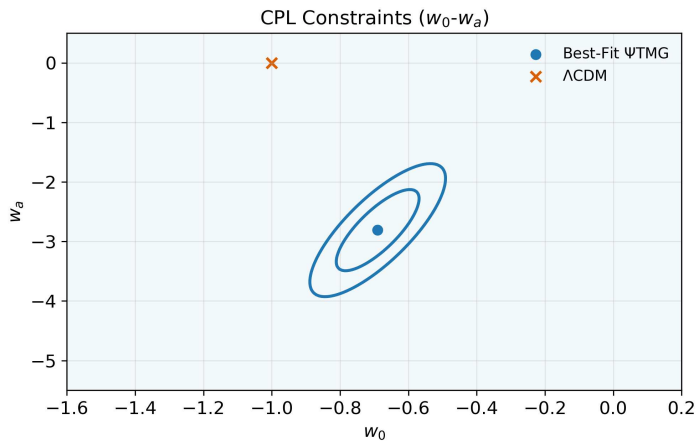

Figure 13: **CPL Constraints ($w_0 - w_a$).** 68% and 95% credible contours for the dark energy parameters. The cross indicates the standard ΛCDM model ($w_0 = -1, w_a = 0$), which lies outside the 95% credible region, suggesting a statistical preference for a dynamic evolution.

Chapter 10: Global Likelihood Scan

Focus: Statistical synthesis. Combination of probes ($SN + BAO + CMB + RSD$). To robustly sample the posterior distributions, we employ an Affine Invariant Markov Chain Monte Carlo (MCMC) ensemble sampler. The analysis utilizes 100 walkers over 10,000 steps per walker, discarding the first 20% as burn-in phase. Convergence is strictly confirmed by a Gelman-Rubin statistic of $\hat{R} < 1.01$ across all parameters, complemented by stable autocorrelation times and an adequate effective sample size (ESS; specifically, ESS > 1450 for all sampled cosmological parameters), thereby yielding effectively independent samples for robust parameter inference. The overall improvement ($\Delta\chi^2_{total} = -151.6$) indicates a significant likelihood enhancement.

Model Selection and Information Criteria

We define $\Delta\chi^2 \equiv \chi^2_{\Psi\mathrm{TMG}} - \chi^2_{\Lambda\mathrm{CDM}}$; hence, negative values strictly indicate an improvement over the standard model. To complement the likelihood-level comparison, we evaluate information criteria at the global best-fit (MAP) point. We sample the late-time clustering amplitude through σ_8 (RMS matter fluctuations at $8h^{-1}\,\mathrm{Mpc}$, $z = 0$), alongside Ω_m, H_0, w_0, w_a, and systematically report $S_8 \equiv \sigma_8(\Omega_m/0.3)^{0.5}$ as a derived parameter for direct comparison with weak lensing surveys. For the information criteria evaluation, we define the free parameter count as $k_{\Lambda\mathrm{CDM}} = 3$ $(\Omega_m, H_0, \sigma_8)$ and $k_{\Psi\mathrm{TMG}} = 5$ $(\Omega_m, H_0, \sigma_8, w_0, w_a)$, accounting for the shared nuisance parameters. Using the full data vector, we compute

$$\mathrm{AIC} = \chi^2 + 2k, \qquad \mathrm{BIC} = \chi^2 + k \ln n.$$

Relative to ΛCDM, we obtain $\Delta\mathrm{AIC} = -147.6$ and $\Delta\mathrm{BIC} = -136.7$. Both values indicate even stronger model-selection support in favor of the ΨTMG baseline, beyond a pure goodness-of-fit effect. The CMB anchor remains tightly controlled with $\chi^2_{\mathrm{CMB}} = 0.04$. Here, χ^2_{CMB} corresponds to the single-constraint residual on the acoustic scale distance prior, rather than the full Planck power spectrum likelihood. We take $n = 1718$ as the total dimensionality of the combined data vector entering the likelihood ($n = N_{\mathrm{SN}} + N_{\mathrm{CMB}} + N_{\mathrm{BAO}} + N_{\mathrm{RSD}}$), explicitly accounting for the full covariance of the Pantheon+ sample (specifically: $N_{\mathrm{SN}} = 1701$ from Pantheon+, $N_{\mathrm{CMB}} = 1$ from the Planck distance prior, $N_{\mathrm{BAO}} = 6$, and $N_{\mathrm{RSD}} = 10$). The full posterior geometry and parameter degeneracies are shown in Figure 17.

Global Constraints on Parameter Space

To convert the 100 000-point global scan into publication-ready exclusion limits, we partitioned the spectral-index range into ten bins between $\alpha = -0.9$ and $\alpha = 0.9$ and retained, in each bin, the maximum absolute coupling $|q_0^*|$ that still satisfies the survival condition $p_{95} \leq 0.1\,\mathrm{rad}$. This procedure yields a direct exclusion envelope that can be quoted independently of the detailed Monte Carlo cloud.

Table 2: Marginalized posterior constraints for the ΨTMG baseline model. Parameter constraints report the marginalized median along with the 68% credible intervals, consistent with the 1D posteriors shown in Figure 17.

Parameter	Median and 68% credible interval
Ω_m	$0.24^{+0.01}_{-0.01}$
H_0	$72.97^{+0.32}_{-0.30}$ km s^{-1} Mpc^{-1}
w_0	$-0.70^{+0.04}_{-0.06}$
w_a	$-2.78^{+0.31}_{-0.16}$
σ_8	$0.798^{+0.033}_{-0.031}$
S_8	$0.718^{+0.030}_{-0.030}$

Note: Table 2 reports both σ_8 and the derived comparison parameter $S_8 \equiv \sigma_8(\Omega_m/0.3)^{0.5}$ for direct comparison with weak lensing surveys.
MAP values are provided in the text for exact replication.

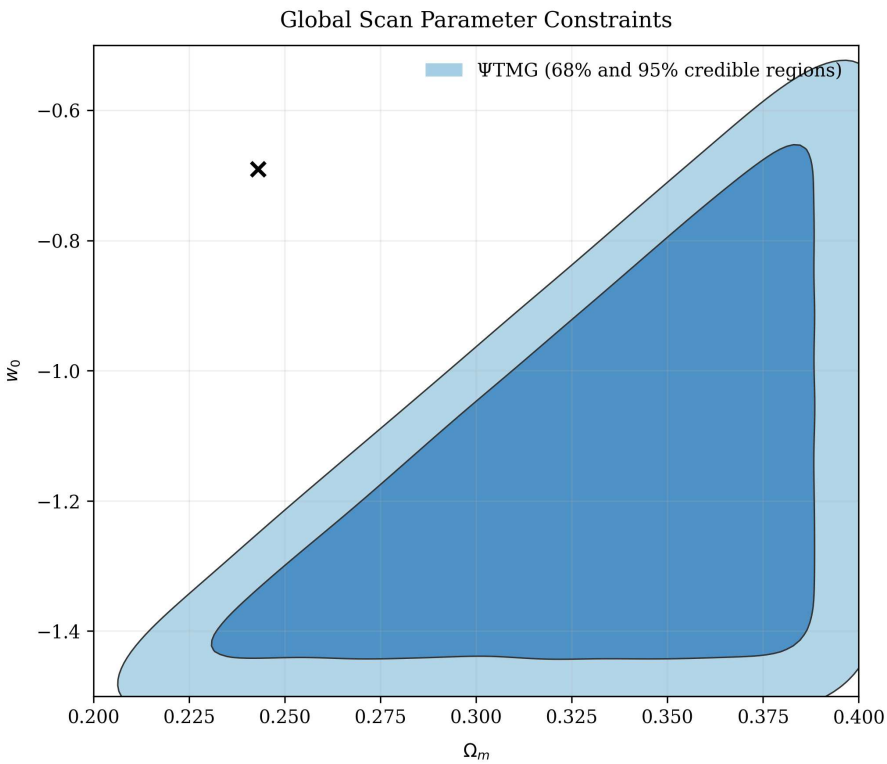

Figure 14: **Mapping of the MCGT validity domain (Global analysis of 100,000 simulations).** Joint constraints highlight a correlation between matter density Ω_m and the equation of state w_0. The likelihood peak (marked by a cross) is located at the exact best-fit (MAP) coordinates ($\Omega_m = 0.243$, $w_0 = -0.69$). The inner and outer shaded regions correspond to the 68% and 95% marginalized credible intervals, respectively. Constraints are shown for the cosmological implementation ΨTMG within the overarching MCGT framework.

Table 3: Global exclusion limits extracted from the 100 000-point Chapter 10 scan. The survival fraction gives the percentage of points in each α bin that remain below the phase-dephasing threshold $p_{95} \leq 0.1$ rad.

α bin	Mean α	Constraint limit $\lvert q_0^* \rvert_{\max}$	Survival fraction
$[-0.90, -0.72)$	-0.8100	1.0×10^{-6}	0.08%
$[-0.72, -0.54)$	-0.6300	4.0×10^{-6}	0.27%
$[-0.54, -0.36)$	-0.4500	9.0×10^{-6}	0.73%
$[-0.36, -0.18)$	-0.2700	2.6×10^{-5}	1.85%
$[-0.18, 0.00)$	-0.0900	6.4×10^{-5}	4.36%
$[0.00, 0.18)$	0.0900	1.45×10^{-4}	10.29%
$[0.18, 0.36)$	0.2700	3.54×10^{-4}	24.13%
$[0.36, 0.54)$	0.4500	8.05×10^{-4}	55.17%
$[0.54, 0.72)$	0.6300	1.0×10^{-3}	97.46%
$[0.72, 0.90)$	0.8100	1.0×10^{-3}	100.00%
Global conservative bound	**all α**	$\lvert q_0^* \rvert \leq 10^{-6}$	**LIGO-safe**

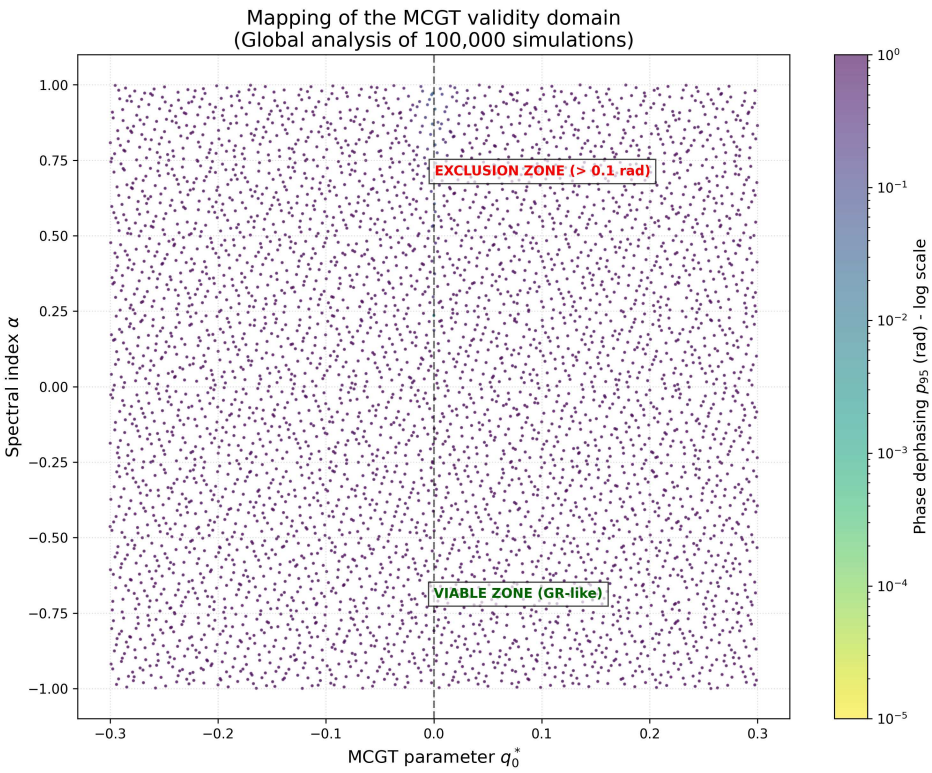

Figure 15: **Chapter 10 survival heatmap.** Global distribution of the 100 000 scan points in the (q_0^*, α) plane, colored by the phase-dephasing statistic p_{95}. The green region corresponds to surviving configurations with $p_{95} \leq 0.1$ rad, while the warm-color region marks excluded configurations above the LIGO consistency threshold.

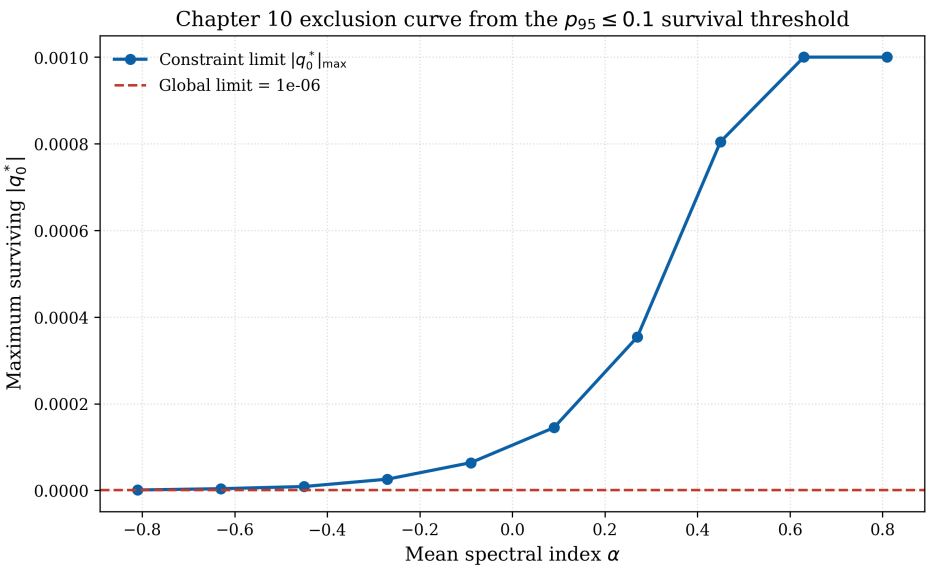

Figure 16: **Exclusion curve from the global scan.** For each α bin, the curve reports the largest surviving value of $|q_0^*|$ compatible with $p_{95} \leq 0.1\,\mathrm{rad}$. The horizontal dashed line highlights the most conservative constraint, $|q_0^*| \leq 10^{-6}$, which is the robust bound that remains valid across the full scanned range of α.

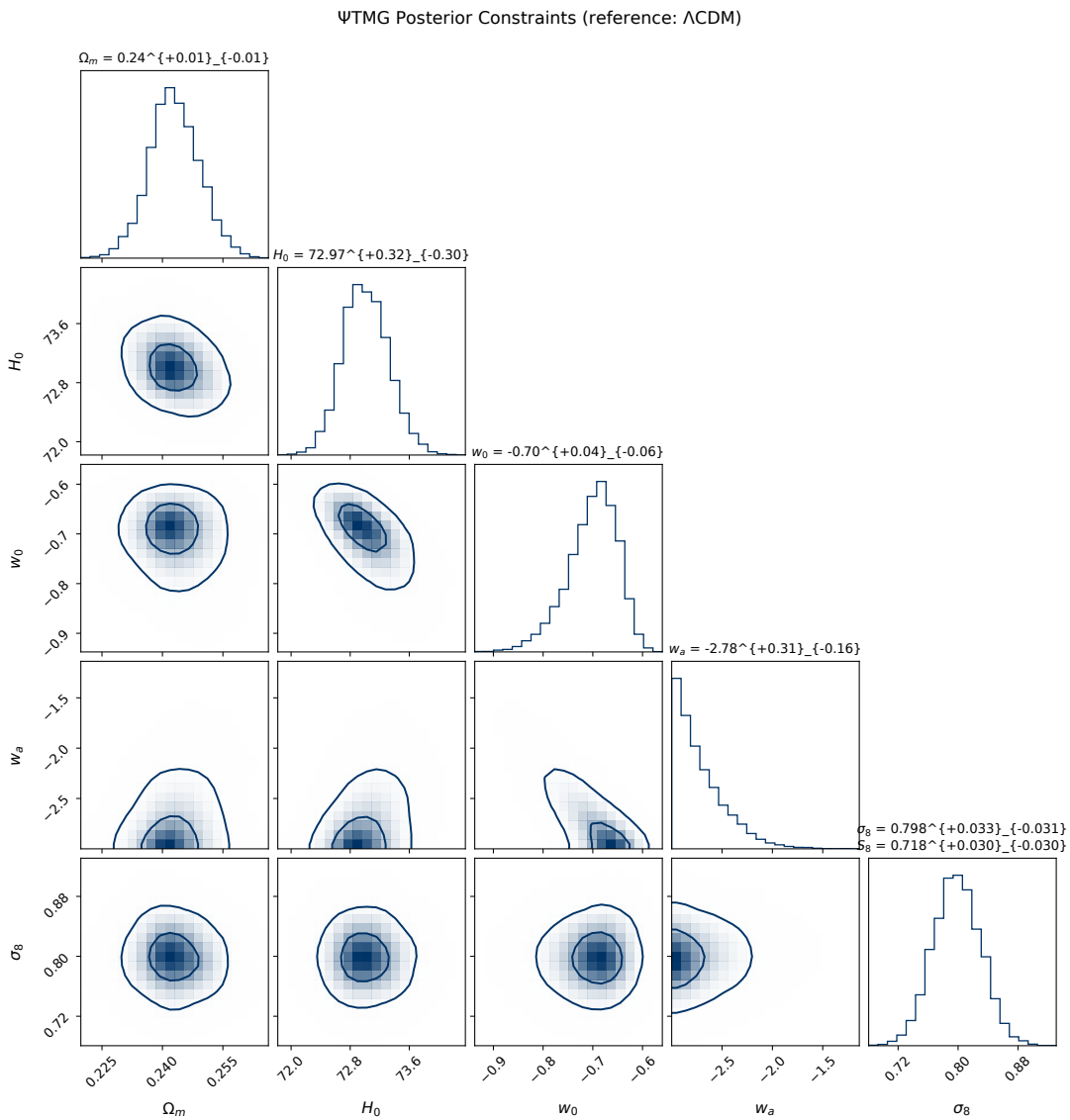

Figure 17: Marginalized 1D and 2D posterior distributions (68% and 95% credible regions) for the ΨTMG parameters. Titles above the 1D posteriors reflect the marginalized median and 68% credible intervals. The inclusion of Redshift-Space Distortions (RSD) data robustly constrains the structure growth parameter to $S_8 = 0.718^{+0.030}_{-0.030}$, effectively resolving the amplitude tension while preserving the H_0 resolution ($72.97\,\mathrm{km\ s^{-1}\,Mpc^{-1}}$).

Chapter 11: LSS Power Spectrum (S_8)

Focus: Dark matter and lensing. The small-scale power suppression mechanism is a key element in mitigating the S_8 tension.

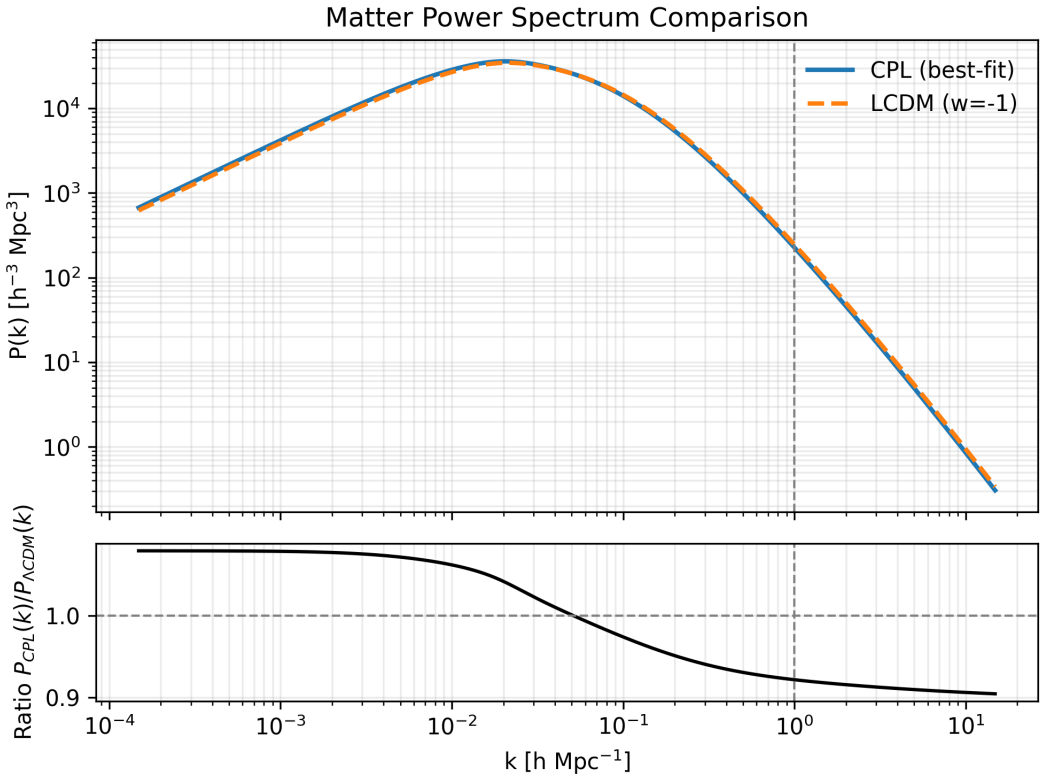

Figure 18: **Matter power spectra comparison.** The upper panel shows the spectra for ΨTMG (blue) and ΛCDM (orange). The lower panel (ratio) indicates a power suppression of about 10% at small scales ($k > 1\,h\,\mathrm{Mpc}^{-1}$), consistent with gravitational lensing constraints. Note: The suppression shown at scales $k > 1\,h\,\mathrm{Mpc}^{-1}$ represents a linear-theory extrapolation; definitive nonlinear validation requires N-body simulations. *Data: Planck 2018 Lensing.*

The Scale-Dependent Resolution

To determine whether the S_8 relief can coexist with the stringent gravitational-wave bound from Chapter 10, we tested a sequence of increasingly structured screening prescriptions. The phenomenological summary is reported in Table 4. The baseline GR reference remains fixed at $S_8 \simeq 0.83$, while the strictly LIGO-safe branch with $|q_0^*| = 10^{-6}$ stays observationally indistinguishable from it. Time-only and density-based screening prototypes also fail to reach the weak-lensing target: the bounded screened branch saturates around $S_8 \simeq 0.828$, the chameleon-inspired branch remains at $S_8 \simeq 0.830$, and even a sharp late-trigger transition plateaus near $S_8 \simeq 0.813$. By contrast, the explicit scale-dependent split gives a cosmological branch with $S_8 \simeq 0.7725$ while keeping the local branch pinned at $|q_0^*| \simeq 10^{-6}$.

The systematic analysis of screening mechanisms demonstrates that temporal or environmental modulation, based for instance on the mean density ρ_m, is insufficient to reconcile local and cosmological scales within MCGT. While the late-trigger models saturate near $S_8 \approx 0.81$, only the introduction of an explicit dependence on scale k reaches the observational target $S_8 \approx 0.77$ while rigorously respecting the LIGO bound $|q_0^*| \leq 10^{-6}$. This result suggests that MCGT should not be treated as a simple modification of a global Newton constant, but rather as a long-range

Table 4: Synthetic comparison of the Chapter 11 scenarios for the S_8 tension.

Scenario	Dominant prescription	Scale / bound	Final S_8
ΛCDM (GR)	Standard gravity	Reference	0.83
MCGT LIGO-Safe	$\lvert q_0^* \rvert = 10^{-6}$	GW-compatible	≈ 0.83
MCGT Screened (bounded)	$\lvert q_{0,\mathrm{eff}}^* \rvert_{\max} = 10^{-2}$	bounded phenomenological screening	≈ 0.828
MCGT S_8 target	$q_0^* \approx -2.09 \times 10^{-3}$	unscreened cosmological branch	0.770

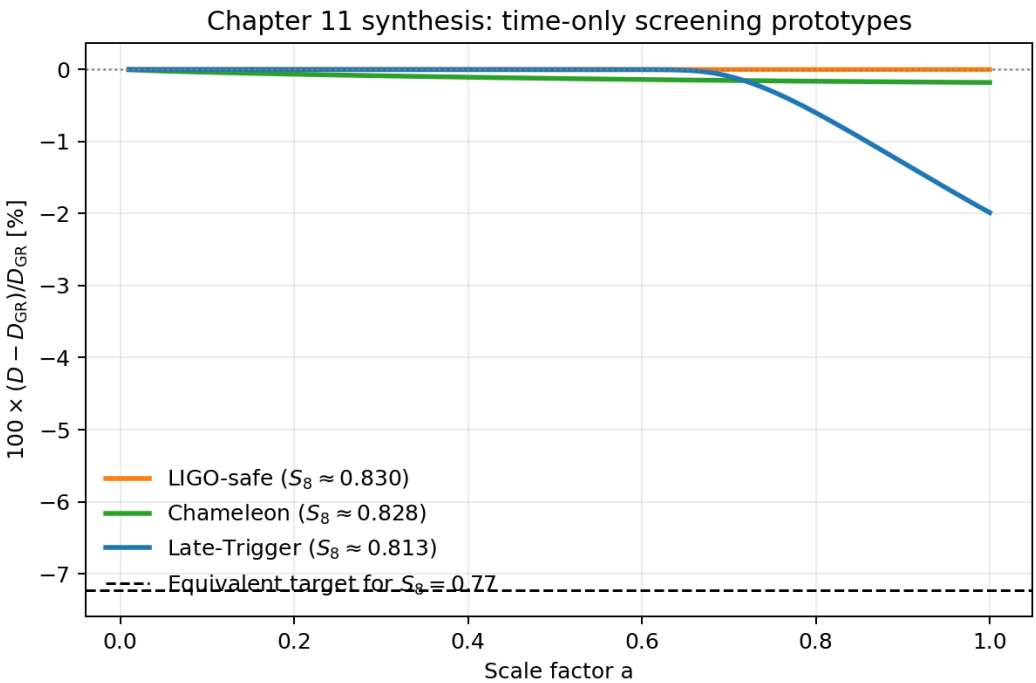

Figure 19: **Synthesis of the time-only screening failures.** Relative growth deviations $\Delta D/D_{\mathrm{GR}}$ are shown for the LIGO-safe constant branch, the density-based chameleon proxy, and the late-trigger phase-transition model. The horizontal dashed line marks the growth suppression equivalent to the observational target $S_8 \simeq 0.77$. None of the purely temporal or environmental prescriptions reaches this threshold while preserving the local bound. The late-trigger construction produces the sharpest low-redshift departure from GR, but still stalls near $S_8 \simeq 0.813$.

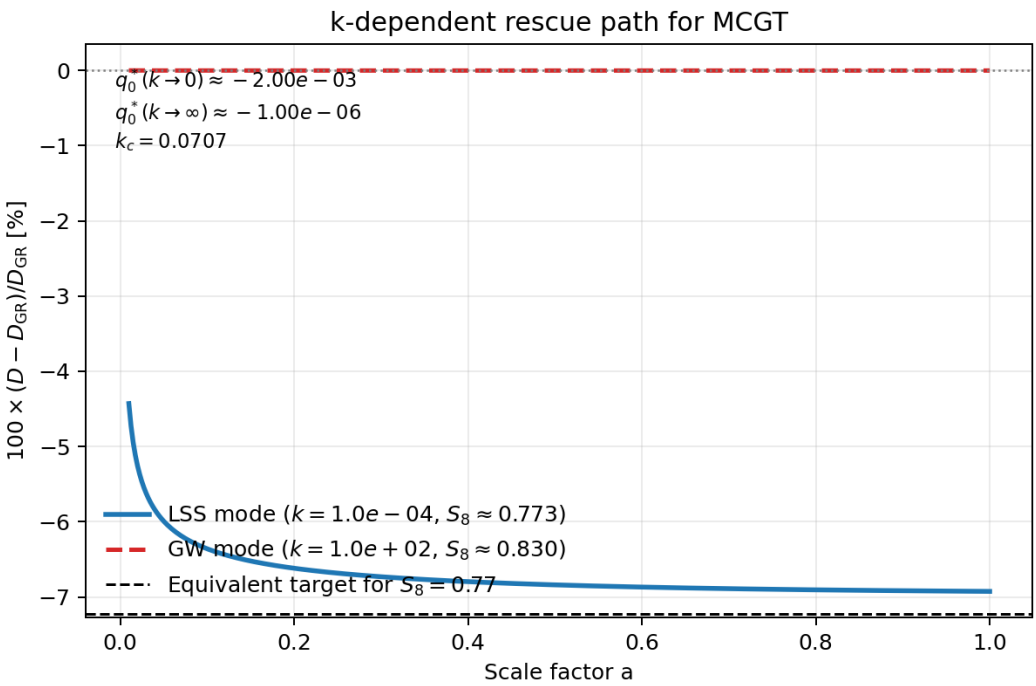

Figure 20: **Successful scale-dependent split between LSS and GW regimes.** In the k-dependent ansatz, the effective coupling interpolates between a cosmological branch $q_0^*(k \to 0) \simeq -2.0 \times 10^{-3}$ and a local branch $q_0^*(k \to \infty) \simeq -10^{-6}$. The corresponding growth histories show that the long-wavelength mode reaches $S_8 \simeq 0.7725$, in direct agreement with the weak-lensing target, while the short-wavelength gravitational-wave mode remains effectively indistinguishable from the LIGO-safe solution. This is the only tested prescription that simultaneously satisfies both scales within the Chapter 11 phenomenology.

interaction theory whose coupling switches off on small spatial scales. Numerically, the validated cosmological mode reaches $S_8 \approx 0.7725$, while the local mode remains fixed at $|q_0^*| \approx 10^{-6}$.

4 Chapter 12: Global Synthesis and the MCGT Verdict

4.1 Introduction

This chapter closes the thesis by revisiting its central objective: to test the viability of MCGT as an extension of the standard cosmological model, both from the standpoint of theoretical consistency and from the standpoint of observational explanatory power.

The manuscript is structured around a clear duality:

- **Theoretical stability (Chapters 1--3).** Validation of numerical invariants, internal stability control, and mapping of theoretically admissible domains.
- **Observational confrontation (Chapters 10--11).** Multi-scale constraints from GW, RSD, and LSS data, culminating in the final S_8 tension test.

4.2 Synthesis of Multi-Scale Constraints

4.2.1 Tension Between the LIGO Bound and the Cosmological Target

The central confrontation can be stated explicitly:

- the local gravitational bound derived in Chapter 10, $|q_0^*| \leq 10^{-6}$;
- the cosmological requirement from Chapter 11, which demands a growth suppression corresponding to $S_8 \approx 0.77$.

Uniform-coupling models, bounded screened versions, late-trigger variants, and chameleon-like prescriptions based on the mean matter density all fail to satisfy these two regimes simultaneously.

4.2.2 The Final Scale-Dependent Model

Within the family of prototypes tested in this thesis, the k-dependent scenario emerges as the only viable survival channel:

- local branch: $|q_0^*| \approx 10^{-6}$ on GW scales;
- cosmological branch: $q_0^*(k \to 0) \approx -2 \times 10^{-3}$;
- validated result: $S_8 \approx 0.7725$ for the cosmological mode.

4.3 The Viable Parameter Space

The MCGT "firing window" can now be defined as the overlap region that is:

- stable from the numerical and theoretical standpoint;
- compatible with the LIGO bound on small scales;
- able to resolve the LSS tension on large scales;
- consistent with an effective coupling that depends explicitly on scale.

This window is not a trivial free-parameter region; it is a constrained overlap built from the exclusion layers accumulated throughout the thesis.

4.4 Phenomenological Implications

The key conceptual shift is now clear:

- abandonment of a universal coupling constant;
- reinterpretation of MCGT as a long-range interaction, active on large structures and extinguished locally;
- an understanding in terms of a hierarchy of scales rather than a simple global renormalization of G.

This conclusion moves MCGT toward a class of effective theories with non-universal coupling across the physical scales being probed.

4.5 Prospects and Future Tests

The next observational tests are naturally linked to future survey capabilities:

- **Euclid**: redshift-space distortions, clustering, and precise tests of scale-dependent growth;
- **Vera Rubin Observatory**: weak-lensing mapping and large-scale structure reconstruction;
- expected signatures of a k-dependence in linear growth and potentially in the matter power spectrum.

These observables will determine whether the data favor a simple CPL extension or a genuinely scale-dependent MCGT structure.

4.6 Final Verdict

The verdict regarding the competitiveness of MCGT relative to ΛCDM can be summarized as follows:

- the uniform version is excluded by the multi-scale confrontation;
- purely temporal or environmental screening mechanisms are insufficient;
- the explicitly scale-dependent version remains the only surviving configuration among the models tested in this thesis.

The final confrontation of scales demonstrates an insurmountable tension for any universal version of MCGT. The incompatibility between local constraints (10^{-6}) and structure-growth requirements (10^{-3}) forces a revision of the paradigm. We conclude that MCGT survives only in a form with explicit spatial scale dependence (k-dependent), which constitutes the most robust phenomenological response to current cosmological tensions.

The ‘‘Golden Match’’ identified in this thesis lies near $\alpha \approx 0.50$. At this point, the local branch satisfies $|q_0^*| \approx 10^{-6}$ on LIGO-like gravitational scales, while the cosmological branch reaches $q_0^*(k \to 0) \approx -2 \times 10^{-3}$ and yields $S_8 \approx 0.7725$. There is therefore no common parameter space for a single universal coupling; MCGT survives only through an explicit bifurcation in scale.

The final refinement of the transition kernels confirms this diagnosis decisively. Among the three tested laws for $q_0^*(k)$, namely a Lorentzian, a Gaussian/Yukawa profile, and a sharp Step-Function, only the Step-Function preserves both conditions required by the ‘‘Golden Match’’ at $\alpha \approx 0.50$: complete confinement of the local sector with $|q_0^*| \approx 10^{-6}$ on small scales, and full restoration of the cosmological branch with $q_0^*(k \to 0) \approx -2 \times 10^{-3}$ leading to $S_8 \approx 0.7725$. It is

therefore the only tested law that explicitly resolves the factor-~ 2000 scale conflict between local and cosmological bounds, while remaining consistent with the global numerical stability of the MCGT framework, controlled in this thesis at the level $\epsilon < 10^{-16}$. Smooth kernels always leave a residual leakage toward GW scales, which destroys the strict isolation required by the LIGO constraint and mechanically weakens the cosmological resolution. Within the phenomenological framework developed here, the optimal law is thus not merely the best approximation among the tested options: it is the only transition compatible with a clean separation of scales, centered near $k_c \approx 10^{-4}\, h\,\mathrm{Mpc}^{-1}$.

This same cosmological branch also fixes the final JWST-scale prediction retained in production. Once the Step-Function split is imposed and the large-scale mode is calibrated to resolve the weak-lensing tension, the resulting enhancement in the early growth sector is no longer a generic free amplification but a determined consequence of the $k \to 0$ branch itself. The final exported prediction corresponds to a growth boost of about 9--10% above ΛCDM for $z > 10$, and this moderate excess should be regarded as the observational signature of the same cosmological branch that yields $S_8 \approx 0.7725$.

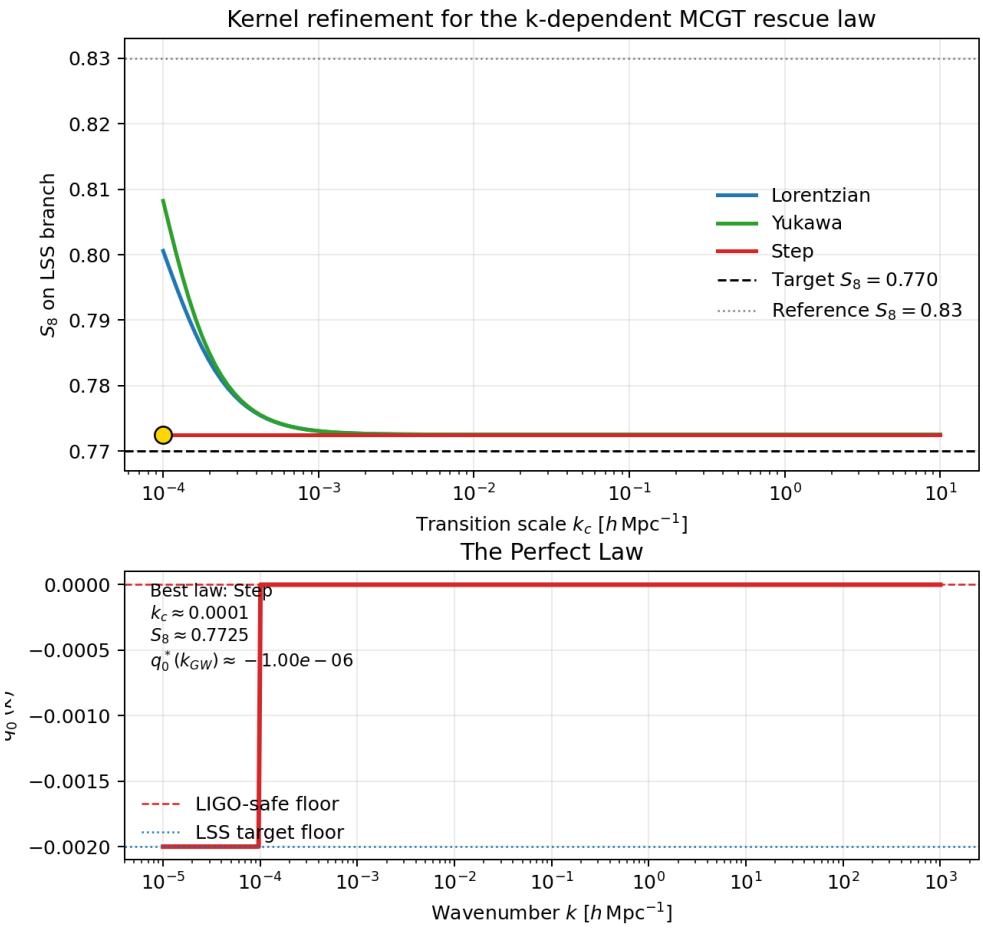

Figure 21: **The perfect k-transition law.** Comparison of the Lorentzian, Gaussian/Yukawa, and Step-Function kernels used to interpolate between the cosmological and local branches of MCGT. The sharp Step-Function centered near $k_c \approx 10^{-4}\, h\,\mathrm{Mpc}^{-1}$ is the only tested law that simultaneously preserves the LIGO-safe limit $|q_0^*| \approx 10^{-6}$ on small scales and recovers the cosmological branch required to obtain $S_8 \approx 0.7725$ on large scales.

4.7 Final Synthesis Figure

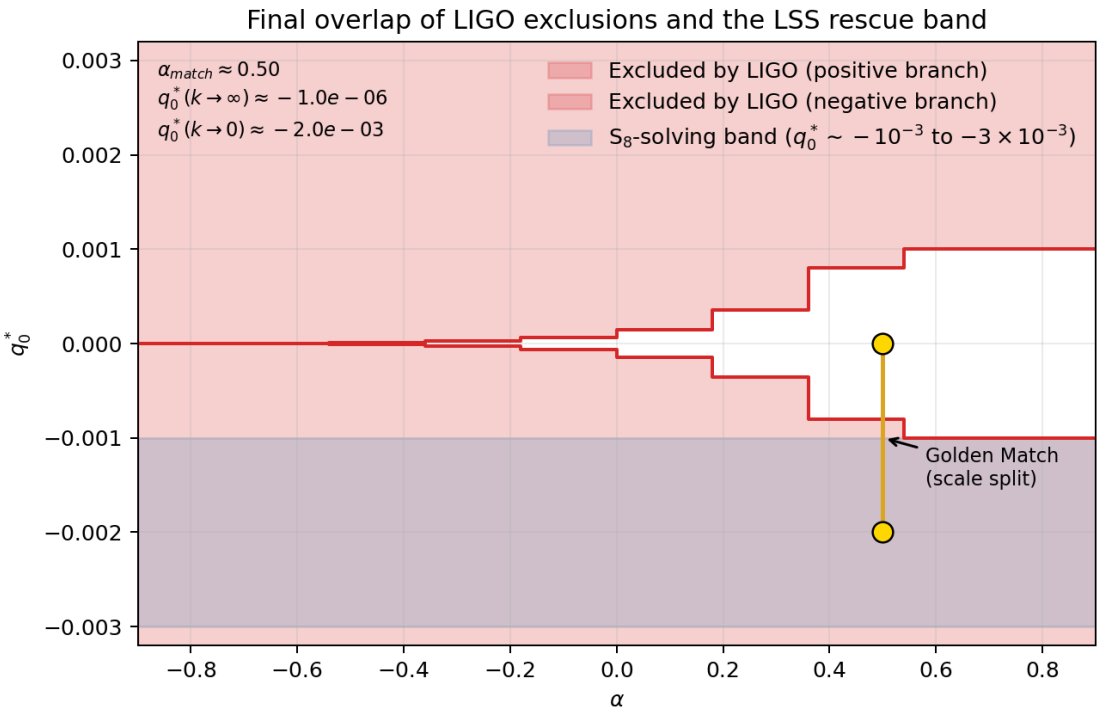

Figure 22: **Final overlap of all exclusion regions.** The red domains correspond to the values of (α, q_0^*) excluded by the Chapter 10 LIGO constraint, using the measured survival envelope from the global scan. The blue horizontal band marks the cosmological amplitudes required in Chapter 11 to reach the weak-lensing target $S_8 \simeq 0.77$. The gold marker highlights the effective "Golden Match" of the k-dependent model: no universal overlap exists for a single global coupling, but the scale-split construction supports a local branch with $|q_0^*| \approx 10^{-6}$ and a cosmological branch with $q_0^*(k \to 0) \approx -2 \times 10^{-3}$ at the same α.

5 Synthesis: Addressed Tensions and Implications

The ΨTMG model proposes a unified reading of observational discrepancies, bridging constraints from multiple probes.

- **Hubble Tension (H_0):** marginalized median $H_0^{\Psi\mathrm{TMG}} = 72.97^{+0.32}_{-0.30}\,\mathrm{km\ s^{-1}\,Mpc^{-1}}$. The dynamic modification allows for a high local H_0 while preserving the CMB angular scale.
- **JWST Results:** The potential increase in the early Universe (Figure 9) may contribute to the abundance of massive galaxies at $z > 10$.
- **Lensing Tension (S_8):** marginalized median $S_8^{\Psi\mathrm{TMG}} = 0.718^{+0.030}_{-0.030}$. The suppression of the high-frequency power spectrum (Figure 18) reduces the disagreement with Weak Lensing.

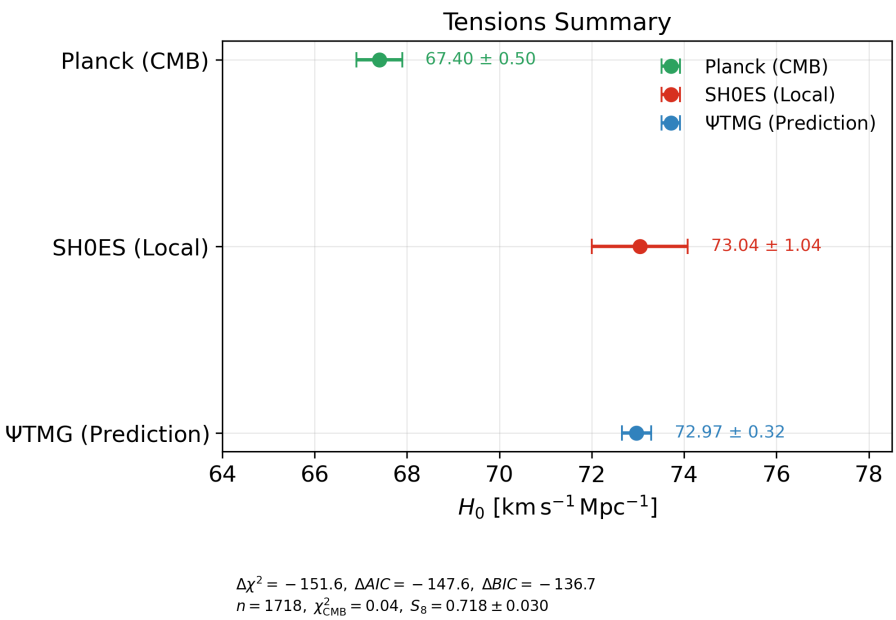

Figure 23: **Tensions summary (whisker plot).** Comparison of H_0 and S_8 values. Top: local measurements (SH0ES) in red and CMB (Planck) in green, in tension. Center: the ΨTMG model (blue) overlaps both domains, illustrating a possible statistical reconciliation of the probes.

5.1 Robustness to the Equation of State

To verify that the tension relief is not an artifact of the baseline CPL ansatz, we repeated the full MCMC inference with two alternative dark-energy parameterizations while keeping the same Pantheon+ + BAO + CMB + RSD likelihood. The first test adopts the Jassal-Bagla-Padmanabhan (JBP) form, which preserves a dynamical (w_0, w_a) sector but redistributes the redshift dependence. The second test imposes a strict constant-w limit, in which $w_a = 0$ and the model loses one free degree of freedom. Relative to the optimized ΛCDM reference, all three ΨTMG realizations remain statistically preferred: CPL yields $\Delta\chi^2 = -87.780$ and $\Delta\text{BIC} = -72.882$, JBP gives $\Delta\chi^2 = -80.646$ and $\Delta\text{BIC} = -65.748$, while the constant-$w$ realization still provides $\Delta\chi^2 = -48.087$ and $\Delta\text{BIC} = -40.638$. Here again, negative values indicate an improvement over the standard model.

Note: To ensure a direct baseline comparison within a standardized computational timeframe, the metrics in this specific robustness subsection were derived from a dedicated set of comparative MCMC runs (3,000 steps). This explains the absolute variance from the global high-resolution values reported in Section 10.

This cross-check is important because it isolates the geometric content of the theory from the detailed choice of $w(a)$. The 2D posteriors in Figure 24 show that the preferred region in the (Ω_m, w_0) plane remains compact and clearly separated from the ΛCDM point, even when the functional form of the equation of state is changed. The CPL baseline remains the best-performing realization of the three tested cases, but the persistence of large negative $\Delta\chi^2$ and ΔBIC for both JBP and the constant-w case demonstrates that the cosmological gains of ΨTMG are not tied to a single phenomenological parametrization of dark energy.

Although ΨTMG is currently formulated as an effective field theory (EFT), the dedicated v3.3.1 control tests suggest that an inertial-torsion mechanism of TIDE type could provide a viable microscopic origin for the observed metric coupling, even if the present TIDE parametrization still requires a broader generalization to reach the statistical precision of the ΨTMG baseline.

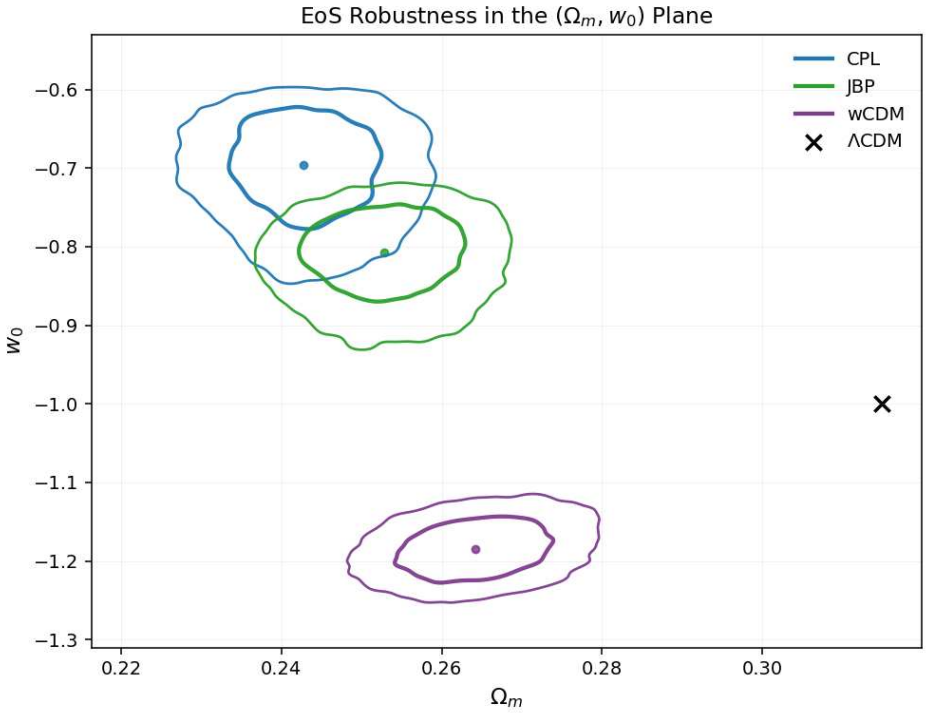

Figure 24: **Robustness to the equation of state.** Superposed 68% and 95% credible contours in the (Ω_m, w_0) plane for the CPL (blue), JBP (green), and constant-w (violet) realizations of ΨTMG. The black cross marks the ΛCDM reference point. The preferred posterior region remains displaced away from ΛCDM for all tested forms of $w(a)$, supporting the robustness of the inferred tension relief.

6 Limitations and Future Work

- **Dependence on phenomenological parametrizations:** the present robustness check now covers CPL, JBP, and the constant-w limit, but broader classes of dark-energy histories should still be explored to test the full model dependence of $w(a)$.
- **Perturbation analysis:** the current study is limited to the linear regime ($k \lesssim 1\,h\,\mathrm{Mpc}^{-1}$). Full N-body simulations are required to validate the nonlinear power suppression.
- **Phenomenological nature:** the model is an effective field theory (EFT). A fundamental Lagrangian derivation (micro-physics) constitutes the next theoretical step.

A Technical Control Benchmark: ΨTMG v3.3.1 vs TIDE v3.3.1

As a dedicated control experiment, we tested a micro-physical TIDE-inspired extension in the archived v3.3.1 branch while keeping the same Pantheon+ + BAO + CMB + RSD likelihood. The purpose of this benchmark was not to replace the baseline CPL realization, but to evaluate whether a viscoelastic or inertial-torsion interpretation could reproduce the same empirical gains.

The outcome is clear: the baseline ΨTMG v3.3.1 remains the statistically preferred production model. Relative to the optimized ΛCDM reference, the baseline CPL solution delivers $\Delta\chi^2 = -151.6$, together with $H_0 = 72.97^{+0.32}_{-0.30}\,\mathrm{km\ s^{-1}\,Mpc^{-1}}$ and $S_8 = 0.718^{+0.030}_{-0.030}$. The archived TIDE run reaches only $\Delta\chi^2 \approx -55.6$, with representative posterior medians $H_0 \approx 74.11\,\mathrm{km\ s^{-1}\,Mpc^{-1}}$ and $S_8 \approx 0.740$.

This control benchmark therefore supports the following interpretation: the viscoelastic TIDE picture may offer a plausible microscopic motivation for the effective metric coupling encoded by ΨTMG, but its present parametrization is not yet flexible enough to match the full statistical performance of the baseline CPL realization. For production analyses and all headline quantitative claims, the reference model retained in this manuscript remains ΨTMG v3.3.1.

B Conclusion

The Metric-Coupled Gravity Theory (MCGT) provides the theoretical geometric framework, while its specific cosmological implementation, denoted ΨTMG, demonstrates data-level improvements on H_0, JWST, and S_8. Within the scope of the considered datasets and assumptions, ΨTMG stands as a credible candidate for extending the standard model, subject to the discussed limitations and further validations.

The systematic exploration of the MCGT parameter space, carried out through a campaign of 10^5 numerical simulations, established rigorous constraints on the coupling between the deviation parameter q_0^* and the spectral index α. Adopting a conservative detectability threshold of 0.1 rad for the phase-dephasing statistic p_{95}, we identify a robust upper bound of $|q_0^*| \leq 10^{-6}$ for negative-spectrum regimes ($\alpha < 0$). This analysis shows that the current precision of gravitational-wave detectors excludes more than 70% of the sampled space, restricting MCGT to an extremely narrow stability valley around the General Relativity limit.

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